Three explicit Diophantine forms of ZFC consistency

Nine nonnegative unknowns, twelve positive unknowns, and twenty-seven unrestricted integer unknowns

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Abstract

We combine the exact blank-input K215 consistency machine, the native tableau/Pell compiler, the four-cell T4 coefficient recipe, the existing 32-variable quartic, and the constructive nine-unknown reduction in the repository's formalization of Jones's 1982 work. This yields a fully specified ordinary polynomial in nine nonnegative integers whose lack of solutions is provably equivalent in ZFC to Con(ZFC). Every coefficient is fixed by a bounded recipe; no recursively enumerable index is left unspecified. Smaller unconditional radical weights reduce the degree of this particular specialization to $161844 \cdot 5^{33} + 484$.

Two further constructions improve the direct arithmetic family. A new four-root combination gives twelve positive unknowns, degree 1609 and 151 binary arithmetic operations. A signed three-root combination gives twenty-seven unrestricted integer unknowns, degree 1004 and 137 operations, or four equations of maximum degree 952 and 131 operations. These are different size tradeoffs. They do not improve the existing 67-operation record and are not asserted globally minimal.

The new arguments are ordinary mathematical proofs, with exact polynomial data, reproducible finite audits and an independent mathematical review. They import the existing compiler and K215 calibration proofs. The archive does not claim a new end-to-end Lean/Rocq build or an internally encoded ZFC derivation.

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1 Results, conventions and comparison

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}_{>0} = \{1, 2, 3, \ldots\}$. A displayed intermediate expression is a polynomial abbreviation unless it is explicitly declared an unknown. All exponents in the final equations are fixed nonnegative integers. Large finite sums, fixed coefficient tables and fixed powers are descriptions of ordinary polynomials, not extra existential quantifiers.

Theorem 1.1 (Main results)**.** *The formulas and fixed coefficient recipes in this report define polynomials $\mathcal{F}_9$, $\mathcal{P}_{12}$ and $\mathcal{P}_{27}$ with integer coefficients such that ZFC proves each of*

$$\begin{aligned}(\nexists \mathbf{x} \in \mathbb{N}^9 : \mathcal{F}_9(\mathbf{x}) = 0) &\iff \mathrm{Con(ZFC)},\\ (\nexists \mathbf{x} \in \mathbb{Z}_{>0}^{12} : \mathcal{P}_{12}(\mathbf{x}) = 0) &\iff \mathrm{Con(ZFC)},\\ (\nexists \mathbf{x} \in \mathbb{Z}^{27} : \mathcal{P}_{27}(\mathbf{x}) = 0) &\iff \mathrm{Con(ZFC)}.\end{aligned}$$

The alternative systems listed below have the same existence predicates. The claim is a provable equivalence, with no assumption of consistency needed for its proof.

Here Con(ZFC) means that there is no finite syntactic ZFC derivation of falsity, with the conventional effective first-order presentation used by the K215 calibration. Equivalent standard proof encodings give the same assertion by their effective translations.

Construction	Domain	Unknowns	Equations	Exact degree	Operations
Nine, new weights	$\mathbb{N}$	9	1	$161844 \cdot 5^{33} + 484$	20,531
Direct twelve	$\mathbb{Z}_{>0}$	12	1	1609	151
Direct twelve	$\mathbb{Z}_{>0}$	12	3	1600	147
Signed, lower weight	$\mathbb{Z}$	27	1	1004	137
Signed, lower weight	$\mathbb{Z}$	27	4	952	131
Signed, full weight	$\mathbb{Z}$	27	1	1116	136
Signed, full weight	$\mathbb{Z}$	27	4	1064	130

For a system, degree means the maximum total degree of a residual. An operation is one binary addition, subtraction or multiplication in the supplied shared arithmetic circuit, including squaring and multiplication by numerals. Equality comparisons, register reuse and supplied fixed coefficient leaves are free, as in the repository's comparisons. The nine-variable circuit expands all fixed powers into multiplications. The counts describe these emitted circuits, not a lower bound on every possible evaluator.

1.1 What becomes simpler

The baseline is the repository at `f0ae4e0991ca44a7bb4e5671d42e33d775e48035`. Its wave-2 comparison [2] reports thirteen positive unknowns for the smallest displayed direct construction, with degree 571 and 84 operations (or degree 515 and 85 operations). Its best listed unrestricted integer count is twenty-nine [9]. The Hilbert-tenth README at this pin explicitly distinguishes the formal existence theorem for nine unknowns from the then-open concrete consistency specialization. Our nine-variable construction fills that fixed-instance specification gap; the general nine-unknown theorem itself is classical [1] and is not new here.

The direct twelve-variable construction deletes another positive coordinate by combining the main Pell root with the three roots already amalgamated in package 09. The twenty-seven-variable construction proves that the three-root amalgamation survives the signed domain of package 14, including two exceptional zero cases that the positive argument could not discard automatically.

The earlier points still matter. A degree-121, fourteen-positive-variable polynomial costs 76 operations; a fifteen-variable polynomial costs 67 operations at degree 155; the 32-variable quartic is a sum of squares of 23 quadratics. There is also a fifteen-variable six-equation system of degree 22 at 61 operations, and a degree-28 system at 60 operations [2]. Thus "simplest" has no single answer without a choice of variable count, degree, circuit size, coefficient height, or ease of proof. This report adds specific points to that comparison rather than claiming one dominates all the others.

1.2 Coefficient size and the domain matter

The basic T4 radix is already

$$B = 2^{5^{18}} = 2^{3\,814\,697\,265\,625}.$$

The binary expansion of $\beta = B - 1$ alone has 3 814 697 265 625 bits, approximately 477 billion bytes. The Jones index constants are much larger. We specify these integers exactly by bounded formulas and supply modular evaluators; we do not pretend that their decimal expansion would be a useful artifact.

The domain of $\mathcal{F}_9$ is essential: its last nonnegative coordinate enforces a strict inequality. Using the same polynomial over all of $\mathbb{Z}^9$ is not justified. Nine strictly positive coordinates are possible by substituting $x_j - 1$ for each nonnegative coordinate, without changing degree. Converting all nine coordinates to unrestricted integers with three-square representations gives another 27-integer form, but retains the huge degree. The separate signed construction below is far smaller under the degree and circuit measures.

2 The logical and compiler interface

2.1 The exact machine and consistency predicate

Let $K215$ be the repository's dense two-symbol, 215-working-state machine, started in state 0 on the blank tape, with a separate halt state 215. The SHA-256 of its literal dense table is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090.`

The source theorem `KEndpoint215.blank_halts_iff_zfc_refutation` has endpoint

$$\mathrm{BlankHalts}(K215) \quad \longleftrightarrow \quad \mathrm{BProv}(\mathrm{ZFCax}_s, [], \bot).$$

The right side is finite syntactic refutability. The ordinary calibration proof explicitly explains how the machine simulation, initialization, source-program equivalence and dominated-calculus calibration compose to a proof formalizable in ZFC [3, 4]. In the formal source the shared calibration passes through ZF refutability and the constructible-choice interpretation relating ZF and ZFC refutations. There is no consistency premise in the endpoint.

The published K215 records report successful Lean and Rocq endpoint builds for this table. They also explicitly distinguish these external proof-assistant results from a derivation encoded in ZFC's own syntax. Our use of "ZFC proves" rests on the ordinary calibration argument formalizable in ZFC, together with the arithmetic proofs here. It is not an inference that a Lean theorem, merely by existing, supplies a ZFC proof object. These builds were not rerun for this report.

2.2 One complete coefficient tuple

All constructions here use the complete T4 tuple of package 11 [5]. Write its four fixed polynomial coefficients as

$$\beta = B - 1, \qquad K_0 = D_C + BD_R, \qquad M_C, \qquad M_F = M_F^{\text{native}} + B - 1.$$

Section 7 defines them without an unbounded search. In particular, the fourth coefficient is the *shifted* mask. The names `Bm1` and `Kconstant` in inherited circuits mean β and K_0, respectively. They do not denote additional inputs. The root `COEFFICIENT_BINDINGS.json` makes these aliases explicit.

The native-window compiler converts a marked cyclic T4 word into arithmetic constraints. The overlap relations pull back the cyclic word to a globally valid plane even for an arbitrary stride: no coprimality or divisibility assumption on that stride is needed. The marked initialization tile then identifies a finite blank-input computation strip. Conversely, every finite halt has sufficiently wide blank margins and an arbitrarily padded halted tail; independent width and height powers of five give the required cyclic presentation. The mask, population and Pell lemmas turn these word constraints into the fixed four-port arithmetic interface [5, 6].

Package 11 proves that this entire T4 tuple satisfies that interface. The low-degree lift and the root eliminations use its interface identities and inequalities, not a particular historical ordering of larger tiles. We therefore substitute the tuple as a whole. Mixing one port from T4 with a historical mask or tile ordering would not be covered by the proof.

2.3 What is imported and what is proved here

The imported mathematical components are: the K215 calibration; the native/Pell compiler including its positive completion; the T4 adapter; the ordinary strong-rank and signed-core arguments; and the constructive quartic-to-nine reduction. Selected exact source files and a source manifest accompany the report. Their transitive historical proof closure is not claimed to be a newly self-contained formal project.

The new work is the explicit normalized quartic and bounded index specialization; smaller unconditional weights in the nine-variable finalizer; the twelve-variable four-root proof; and the signed twenty-seven-variable proof. Algebraic generators, exact identity checks, degree certificates and modular evaluations support these constructions. A finite check cannot replace a compiler theorem about arbitrary runs or the universal arithmetic inequalities used below.

For any of the constructions the proof chain is

a solution in the stated domain $\longleftrightarrow$ a finite blank halt of $K215$ $\longleftrightarrow$ a finite ZFC refutation.

Negating this chain is exactly the required consistency equivalence.

3 A concrete normalized quartic and its bounded index

3.1 The twenty-three quadratic equations

We use package 09's ordinary-strong 32-variable lift [7]. Its unknowns, in the exact coefficient-file order, are

$$\begin{aligned}\mathbf{v} = (&J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,\\ &N, E, P, c, a, X, D, C_2, G, L, F_2, V, g, I, B_{\text{aux}}, \Delta) \ \in \mathbb{Z}_{>0}^{32}.\end{aligned}$$

The affine abbreviations are

$$q = \beta J + 1, \qquad k = \eta + \zeta, \qquad H = 4a + 3, \qquad C_w = q - F - \alpha.$$

The residuals r_j, all of total degree at most two in these unknowns, are the following. The four fixed ports have degree zero.

j	$r_j(\mathbf{v})$
1	$L - q^2 + 1$
2	$N - q(L + 1)$
3	$X - wq$
4	$a - E - sN$
5	$E - X(a - E)$
6	$c - k(a - E) - \eta$
7	$P - E(c - \eta) - k$
8	$\tau^2 - P(P - k) - 1$
9	$D - X - ac - \gamma H$
10	$\Delta - a^2 - H$
11	$C_2 - c^2$
12	$D^2 - \Delta C_2 - 1$
13	$k - jc + V - 1 - hE$
14	$G - (q - 1)(q - F) - \alpha$
15	$jc - V - LG - (M_C + qM_F)J$
16	$(K_0 + w)C_w + q - F - z(q - 1) - 1$
17	$F_2 - f^2$
18	$V - of + c$
19	$g - y^2 + V^2$
20	$I - iC_2$
21	$B_{\text{aux}} - I^2$
22	$B_{\text{aux}} - \Delta(F_2 - 1)$
23	$y^2 - B_{\text{aux}} g - 1$

This is already a useful low-degree answer to the original question: the simultaneous equations $r_1 = \cdots = r_{23} = 0$, or the single quartic $\sum r_j^2 = 0$, have a positive solution exactly when K215 halts. It is an inherited construction, included to make the nine-variable specialization fully determined.

For orientation, the first six rows recover $N = q^3$, $Y = a - E = sq^3$, $X = wq$, $E = XY$, and $c = kY + \eta$. Rows 7–8 give the factored first Pell norm. Rows 9–12 give the main Pell norm. Rows 13–16 encode the packed index and transport equation. The remaining rows give the ordinary strong relation $(ic^2)^2 = \Delta(f^2 - 1)$ and the auxiliary norm. This ordinary strong relation is deliberately distinguished from $f^2 - \Delta i^2 c^4 = 1$, used by the normalized direct constructions. The ordinary strong-rank lemma supplies soundness; positive canonical witnesses supply every lifted register, including the positive gap $g = y^2 - V^2$. These projection and extension claims are proved in the bundled low-degree proof, not inferred merely from the number of rows.

3.2 Normalization is explicit

Introduce 32 nonnegative coordinates $n_1, \ldots, n_{32}$ and a dummy input x which does not occur in the polynomial. Define

$$P_{32}(x, \mathbf{n}) = \sum_{j=1}^{23} r_j(\mathbf{n} + \mathbf{1})^2. \tag{1}$$

Because a sum of integer squares is zero exactly when each summand vanishes, its nonnegative zeros are precisely the positive solutions of the quadratic system. It has degree exactly four: the coefficient of n_τ^4 is one.

At the all-ones positive tuple, the fifth residual is $1 - 1(1-1) = 1$. Consequently

$$P_{32}(x, 0, \dots, 0) \geq 1 \quad \text{for every } x \in \mathbb{N}.$$

This discharges the normalization hypothesis of the Jones construction without an extra variable or a case split. The exact upstream sparse file has SHA-256

`149eacd479896a113d869c5b74163efa4dab2cabaca8b581a6405cc9f580ca60`.

3.3 Every index constant is specified

Write $\nu = 32$, $L = 5^{33}$, and let $\alpha = (\alpha_0, \dots, \alpha_{32}) \in \mathbb{N}^{33}$ with $|\alpha| \leq 4$. Define

$$p_\alpha = [x^{\alpha_0} n_1^{\alpha_1} \cdots n_{32}^{\alpha_{32}}] P_{32},$$
$$c_\alpha = (4 - |\alpha|)! \prod_{j=0}^{32} \alpha_j!\, p_\alpha, \qquad e(\alpha) = \sum_{j=0}^{32} \alpha_j 5^j.$$

The slack factorial is essential. Merely using the ordinary monomial coefficient p_α would not satisfy the source coding identity. There are $\binom{37}{4} = 66\,045$ eligible multi-indices. The shifted quartic has 346 nonzero witness-coefficient polynomials, comprising 508 terms when the four fixed ports are retained symbolically. All nonzero records have $\alpha_0 = 0$, but the baseline sum below still includes the multi-indices with a nonzero input exponent.

Here is a deterministic bound with no coefficient search. For each raw residual, first expand the affine abbreviations in the table and write

$$r_j(\mathbf{v}) = \sum_{a,b} d_{j,a,b}\, \mathbf{v}^a (\beta, K_0, M_C, M_F)^b, \qquad d_{j,a,b} \in \mathbb{Z}.$$

Set

$$\rho_j = \sum_{a,b} |d_{j,a,b}|\, 2^{|a|} (\beta, K_0, M_C, M_F)^b, \qquad s_\star = 2 + 24 \sum_{j=1}^{23} \rho_j^2, \qquad z_\star = 2^{s_\star}.$$

All four ports are positive integers. The factor $2^{|a|}$ bounds the coefficient ℓ^1 norm after the shift $\mathbf{v} = \mathbf{n} + 1$; the norm is submultiplicative. Each factorial multiplier is at most $4! = 24$. Therefore

$$\sum_{|\alpha| \leq 4} |c_\alpha| \leq 24 \sum_j \rho_j^2 = s_\star - 2, \qquad |c_\alpha| + 1 < 2^{s_\star} = z_\star.$$

The explicitly expanded bound is only a twenty-term polynomial:

$$\begin{aligned} s_\star = {} & 79250 + 960M_F + 96M_F^2 + 960M_C + 192M_C M_F + 96M_C^2 \\ & + 2880K_0 + 600K_0^2 + 9504\beta + 1920\beta M_F + 384\beta M_F^2 + 384\beta M_C M_F \\ & + 3552\beta K_0 + 480\beta K_0^2 + 6048\beta^2 + 384\beta^2 M_F^2 + 960\beta^2 K_0 + 96\beta^2 K_0^2 \\ & + 1920\beta^3 + 768\beta^4. \end{aligned}$$

Finally define the fixed digit polynomial and constants

$$e_\star(T) = \sum_{|\alpha| \leq 4} (z_\star + c_\alpha) T^{L - e(\alpha)},$$
$$u_\star = \sum_{j=1}^{32} (2z_\star)^{5^j}, \qquad y_\star = e_\star(2z_\star).$$

All exponents are nonnegative since $e(\alpha) \le 4 \cdot 5^{32} < L$. The coefficient bound makes every digit positive; in particular $z_\star, u_\star, y_\star$ are natural numbers. These definitions satisfy the source predicate `Index 32` P_{32} $z_\star$ $u_\star$ $y_\star$ exactly [10]. The value $u_\star$ is part of the proof interface but is not a free leaf of the final polynomial.

4 An explicit equation in nine nonnegative integers

The classical nine-unknown construction can be applied directly to the concrete quartic P_{32} above. This does not require an unspecified index of a recursively enumerable set. Every constant in what follows is supplied by the bounded coefficient recipe for P_{32}. Write $z_\star, y_\star$ for its chosen index constants, put $\nu = 32$ and $L = 5^{\nu+1} = 5^{33}$, and write

$$e_\star(T) = \sum_{\substack{\alpha \in \mathbb{N}^{\nu+1} \\ |\alpha| \le 4}} (z_\star + c_\alpha) T^{L - \sum_{r=0}^{\nu} \alpha_r 5^r}$$

for the fixed univariate digit polynomial constructed from the weighted coefficients c_α of P_{32}. Its exponent bounds are fixed integers. In particular, evaluating $e_\star$ does not introduce any new existential unknown.

4.1 The nine unknowns and their polynomial expressions

The unknowns are exactly

$$(\varepsilon, g_0, h_0, s_0, w_0, \phi, i_0, j_0, n) \in \mathbb{N}^9.$$

Set $g = g_0 + 1$, $h = h_0 + 1$, $s = s_0 + 1$, $w = w_0 + 1$, $i = i_0 + 1$, and $j = j_0 + 1$. All symbols defined below are polynomial abbreviations, not additional unknowns. First put

$$\begin{aligned}
b &= y_\star + 1 + \varepsilon, & \mathcal{B} &= b^5, & q &= \mathcal{B}^L, \\
\ell &= \sum_{r=1}^{\nu} \mathcal{B}^{5^r}, & \lambda &= \sum_{r=0}^{4L-1} \mathcal{B}^r, & e &= e_\star(\mathcal{B}), \\
\theta &= \mathcal{B} - 2z_\star, & N &= q^{16}, & G &= 32b\mathcal{B}^{5^\nu}.
\end{aligned}$$

The packed code is the following explicit polynomial:

$$\begin{aligned}
R = {} & \Big[g + eq^3 + \ell q^5 + (2(e - z_\star\lambda)(1 + \mathcal{B} + g)^4 + \lambda\mathcal{B} + \lambda\mathcal{B}q^4)q^7\Big](N^2 - N) \\
& + \Big[q^3 - b\ell + \ell + \theta\lambda q^3 + (\mathcal{B} - 2)q^8\Big](N^2 - 1). \qquad (2)
\end{aligned}$$

Next define

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M &= RY, \\
P &= 2M^2 U, & K &= R + 1 + h(P - 1), & A &= M(U + 1), \\
B_1 &= 2R + 1, & W_1 &= bw, & C &= B_1 + W_1 + \phi, \\
J &= 4A - 5, & Q &= 3W_1 C - 2(W_1^2 - 1), & D &= (A^2 - 1)C^2 + 1, \\
E &= iJDC^2, & F &= (A^2 - 1)E^2 + 1, & G_1 &= A + F(CD + 1 - A), \\
H &= B_1 + jC, & I &= (G_1^2 - 1)H^2 + 1.
\end{aligned}$$

The four arithmetic conditions in the Jones ratio reduction use

$$\begin{aligned}
a_0 &= (P^2 - 1)K^2 + 1, & a_1 &= DFI, \\
d &= (JF)^2, & c_{\mathrm{div}} &= (QF + J(H - C))^2, \\
m &= K^2 G - 4(C - KY)^2 G - 8gK^2.
\end{aligned}$$

Here c_{div} is a dividend, and is unrelated to the coefficient symbols c_α.

4.2 Smaller unconditional radical weights

Lemma 4.1 (Polynomial majorant for every Pell radicand)**.** *For arbitrary real a, b, set $\mathcal{M}(a,b) = a^2+b^2+1$. Then*

$$\sqrt{|(a^2-1)b^2+1|} \le \mathcal{M}(a,b).$$

Proof. The triangle inequality gives $|(a^2-1)b^2+1| \le (a^2+1)b^2+1$, while

$$(a^2+b^2+1)^2 - \big((a^2+1)b^2+1\big) = a^4+b^4+a^2b^2+2a^2+b^2 \ge 0.$$

Both sides of the desired square-root bound are nonnegative. □

Consequently the following integer polynomial weights are admissible on *every* integer assignment of the intermediate expressions:

$$\begin{aligned} V_0 &= P^2+K^2+2, \\ V_1 &= (A^2+C^2+1)(A^2+E^2+1)(G_1^2+H^2+1)+1. \end{aligned} \tag{3}$$

Indeed, $V_0, V_1 \ge 2$ and $\sqrt{|a_r|} \le V_r - 1$ for $r = 0, 1$. No sign conclusion about R, no square condition, and no positivity conclusion from the margin m is used in this assertion. This avoids any circular appeal to conditions that the final equation is supposed to enforce.

Put

$$\begin{aligned} \Omega &= d(2m-1), \\ \Theta &= dn + c_{\mathrm{div}} - \Omega(c_{\mathrm{div}} + V_0V_1), \end{aligned}$$

$$\boxed{\mathcal{F}_9 = \big(\Theta^2 - \Omega^2(a_0 + V_0^2 a_1)\big)^2 - 4\Omega^4 V_0^2 a_0 a_1.} \tag{4}$$

The requested equation is $\mathcal{F}_9 = 0$ over $\mathbb{N}^9$. Only addition, subtraction, multiplication and powers with fixed nonnegative integer exponents occur. The displayed formula is therefore an ordinary integer-coefficient polynomial, not an exponential Diophantine equation or a formula containing hidden quantified variables.

4.3 Why the polynomial has exactly the required zero-existence predicate

Lemma 4.2 (The finalizer enforces the four ratio conditions)**.** *For every integer tuple of the intermediate variables,*

$$\exists n \in \mathbb{N}\ \mathcal{F}_9 = 0 \quad\Longleftrightarrow\quad a_0, a_1 \text{ are integer squares}, \quad JF \mid QF + J(H-C), \quad m > 0.$$

Proof. First $J = 4A - 5 \neq 0$. Also $F \neq 0$: if $(A^2-1)E^2+1 = 0$, then the integer equation $(1-A^2)E^2 = 1$ forces $A = 0$ and $E = \pm 1$. But $E = iJDC^2$ would then be divisible by 5, a contradiction. Thus $d = (JF)^2 > 0$, and $c_{\mathrm{div}} \ge 0$.

For arbitrary formal square roots $r_0^2 = a_0$ and $r_1^2 = a_1$, direct multiplication gives the polynomial identity

$$\mathcal{F}_9 = \prod_{\sigma,\tau\in\{-1,1\}} (\Theta - \Omega(\sigma r_0 + \tau V_0 r_1)).$$

This identity eliminates the radicals completely; the left side is the ordinary polynomial in (4). Suppose it vanishes. Because $d > 0$ and $2m - 1 \neq 0$, we have $\Omega \neq 0$. Choose complex square roots and absorb the signs of a vanishing factor into them. Then

$$s_* := \Theta/\Omega = r_0 + V_0 r_1 \in \mathbb{Q}.$$

If $s_* \neq 0$, the identity $2s_*r_0 = s_*^2 + a_0 - V_0^2 a_1$ shows that both roots are rational, and hence integral because their squares are integers. If $s_* = 0$, then $|r_1| = |r_0|/V_0 \leq (V_0 - 1)/V_0 < 1$. Consequently the integer a_1 has absolute value less than one, so $a_1 = a_0 = 0$. This also gives integral roots. This argument covers signed radicands and repeated square classes without an independence assumption.

Write $Z = c_{\mathrm{div}} + V_0V_1 + r_0 + V_0r_1$. The root bounds give $Z \geq c_{\mathrm{div}} + 1 > 0$. The vanishing factor is exactly

$$dn + c_{\mathrm{div}} = d(2m-1)Z.$$

It forces $d \mid c_{\mathrm{div}}$ and $2m - 1 > 0$, hence $m > 0$. Conversely, if both radicands are integer squares, $d \mid c_{\mathrm{div}}$ and $m > 0$, choose either signs of their integer roots and put

$$n = (2m-1)Z - c_{\mathrm{div}}/d.$$

This is an integer, and $Z \geq c_{\mathrm{div}} + 1$ and $d \geq 1$ imply $n \geq 1$. Its factor vanishes, so $\mathcal{F}_9 = 0$. This is the two-radical case of refined relation combining, proved here directly. Finally, for integers $u \neq 0, v$, $u^2 \mid v^2$ is equivalent to $u \mid v$. Apply this to $u = JF$, $v = QF + J(H - C)$. □

Theorem 4.3 (Concrete nine-variable consistency equation)**.** *With the complete fixed coefficient tuple and the bounded index recipe specified above, ZFC proves*

$$(\nexists \mathbf{x} \in \mathbb{N}^9 : \mathcal{F}_9(\mathbf{x}) = 0) \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

Proof. The hypotheses of the repository theorem `Jones1982.Nine.wset_iff_nine` are all concrete here: $\nu = 32 \geq 1$, P_{32} has degree four, its normalization follows from the fifth residual at the all-ones tuple, the stated index recipe satisfies `Index`, and the fixed input is $1 > 0$.

More explicitly, `Nine.reduced_of_mem` and `Nine.mem_of_reduced` convert a zero of P_{32} into, and back from, the two-witness packed central-binomial criterion. The theorems `Nine.ratio_complete` and `Nine.ratio_sound` convert that criterion into exactly the four conditions of Lemma 4.2. Their construction uses the eight nonnegative coordinates $(\varepsilon, g_0, h_0, s_0, w_0, \phi, i_0, j_0)$, with the six displayed positive shifts. The original Lean construction combines those conditions with the weights $a_r^2 + 2$; replacing those weights by (3) preserves existence of the final witness n by Lemma 4.2. This is an equivalence of existential zero sets; the same numerical value of n need not work for both choices of weights.

It follows that $\mathcal{F}_9$ has a nonnegative solution exactly when P_{32} does, hence exactly when the specified blank-input K215 machine halts. The established K215 calibration identifies the latter with a finite ZFC derivation of falsity. Negating these equivalences proves the claim. The argument uses the mathematical native/Pell compiler and calibration results already identified for P_{32}; it does not promote finite numerical tests to semantic proofs. □

4.4 Exact degree and a fully expanded binary circuit

The polynomial in (4), with input 1 already fixed, has exact total degree

$$\boxed{161844\,5^{33} + 484 = 18\,841\,121\,345\,758\,438\,110\,351\,562\,984.}$$

The following ledger gives selected exact degrees; all compiler and index constants are coefficients.

Expression	Degree	Expression	Degree
R	$235L$	K	$950L + 4$
A	$555L + 2$	D	$1580L + 4$
E	$2605L + 7$	F	$6320L + 18$
G_1	$8135L + 22$	I	$16740L + 46$
a_0	$3800L + 14$	a_1	$24640L + 68$
d	$13750L + 40$	m	$2221L + 11$
V_0	$1900L + 8$	V_1	$22590L + 62$
Ω	$15971L + 51$	Θ	$40461L + 121$

No degree estimate assumes a solution. For a direct certificate of exactness, specialize all nine unknowns to t while leaving every fixed constant symbolic. The leading coefficient of Θ is 2^{82}, the term Θ^4 has strictly larger degree than the other terms in the final norm, and the leading coefficient of $\mathcal{F}_9(t, \dots, t)$ is 2^{328}. This is a nonzero integer independent of the coefficient tuple. The accompanying symbolic leading-term audit verifies every binary gate and records no unresolved leading cancellation.

The generated arithmetic DAG has exactly 20 531 live binary operations: 664 additions, 22 subtractions and 19 845 multiplications. These counts expand every fixed exponent into binary multiplications and charge squaring and multiplication by a fixed coefficient. They do not charge construction of the giant fixed integers, consistently with the inherited fixed-coefficient operation model. There are nine variable leaves and 348 named fixed leaves: $z_\star$, $y_\star$, and the 346 nonzero symbolic weighted coefficients of the shifted quartic. Those 348 leaves are completely defined constants, not existential parameters.

The long digit-polynomial sum is generated efficiently. If $T = \mathcal{B}$ and $r = 5^\nu$, then

$$\sum_{|\alpha| \le 4} T^{L - \sum \alpha_j 5^j} = T^r h_4(T^r, T^{r-1}, T^{r-5}, \dots, T^{r-5^\nu}),$$

where h_4 is the complete homogeneous symmetric polynomial of degree four. Its standard ascending recurrence uses only addition and multiplication. Thus $e_\star(T)$ is $z_\star$ times this baseline plus the 346 sparse coefficient corrections. The geometric sum defining λ likewise has a binary doubling circuit without division.

For comparison, keeping the original weights $a_0^2 + 2, a_1^2 + 2$ gives a nine-variable equation of degree $291404L + 860$ with 20 524 operations. The unconditional majorants reduce the degree by approximately 44.46% for seven extra additions. The large increase in degree and arithmetic work relative to the small Pell equations is an explicit cost of reducing the number of unknowns; no simultaneous improvement in every size measure is claimed.

5 A direct construction with twelve positive unknowns

The nine-variable compiler and the smaller arithmetic circuits optimize different measures. Here is a further direct reduction of the native/Pell family: twelve strictly positive unknowns, three equations of maximum degree 1600, or one equation of degree 1609. The emitted single-equation evaluator has 151 binary operations (97 multiplications and 54 additions or subtractions); the system uses 147. Thus it improves the thirteen-variable direct construction in variable count at the cost of degree and arithmetic size. All four coefficient ports below are bound together to the fixed T4 tuple specified earlier. No ports from different compiler recipes are mixed.

This section uses two established interface facts from the source packages. Package 09 proves its weak-sign bootstrap and weighted-root inequality for the native four-port interface. Package 14 proves the signed-core soundness argument, including recovery of the main-root sign. In that soundness argument the old strong multiplier need only be nonzero and the scale

w positive. Oddness of that multiplier and squareness of w are sufficient conditions used in its particular unrestricted-integer parameterization, and are not additional hypotheses on the positive construction below. These arithmetic interface arguments transfer with the whole T4 tuple.

5.1 The polynomial

The unknowns are

$$J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t \in \mathbb{Z}_{>0}.$$

All other symbols in this subsection abbreviate polynomials. Write $\beta = B-1$ and let K_0, M_C, M_F be the other three fixed ports, with the shifted M_F convention. Define

$$\begin{aligned} q &= \beta J + 1, & C_w &= q - F - \alpha, & X &= wq, & Y &= sq^3, \\ E &= XY, & k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, \\ H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1. \end{aligned}$$

$$\begin{aligned} R &= ((q-1)(q-F) + \alpha)(q^2-1) + (M_C + qM_F)J, \\ N_k &= k - hE - R, & N_t &= (K_0 + w)C_w + q - F - z(q-1), \\ L_0 &= EkY, & A &= 1 + L_0(L_0 + k). \end{aligned}$$

The new strong multiplier is the old multiplier restricted to Hi:

$$\begin{aligned} I &= Hic^2, & D &= 1 + \Delta I^2, & Q &= \Delta^2 I^2, & d &= Q - 1, \\ v &= cT, & z_0 &= c + RD, & e &= Q(Dv^2 + z_0^2) - 1, & b &= 2Qvz_0, \\ M &= 2(D + v^2), & G &= 1 + \Delta c^2. \end{aligned}$$

For the separating weight put

$$\begin{aligned} T_* &= L_0 + k + 1, & F_* &= (a+2)I + 1, \\ Y_* &= 2(vF_* + z_0), & U_* &= MT_* + Y_*, \\ \ell &= 1 + HU_*, & S &= Ht + \ell(ac + X), & W &= HM. \end{aligned}$$

The following fixed computation defines a polynomial without introducing an unknown. Work formally modulo $r^2 - G$ and form

$$\begin{aligned} Z &= S - \ell r, \\ U &= d(Z^2 + W^2 A) - H^2 e, & V &= 2dWZ, \\ F_0 &= U^2 + AV^2 - H^4 b^2 D, & F_1 &= 2UV, \\ F_0^2 - AF_1^2 &= E_0 + E_1 r \pmod{r^2 - G}, \\ \mathcal{R}_{12} &= E_0^2 - GE_1^2. \end{aligned}$$

Every reduction replaces r^2 by G, so E_0, E_1 are ordinary integer polynomials in the twelve supplied unknowns. This is a finite arithmetic recipe, not an existential use of r. The supplied JSON circuit performs it with coefficient pairs; in particular it evaluates $H^4 b^2 D$ as $4H^4(QDv^2)(Qz_0^2)$.

Our system and single equation are respectively

$$N_k = 1, \qquad N_t = 1, \qquad \mathcal{R}_{12} = 0, \tag{5}$$

and

$$\boxed{\mathcal{P}_{12} = N_k N_t (1 + \mathcal{R}_{12}) - 1 = 0.} \tag{6}$$

Theorem 5.1. *With the stated fixed coefficient binding, each of* (5) *and* (6) *has a positive integer solution exactly when K215 halts on blank tape. Consequently ZFC proves that its absence of positive integer solutions is equivalent to* Con(ZFC).

5.2 Soundness: separating and recovering four roots

We prove soundness under the more general assumptions $N_k = N_t = \varepsilon \in \{1, -1\}$ and $\mathcal{R}_{12} = 0$; this also covers the product finalizer. The inherited weak-sign bootstrap gives

$$q \geq 16, \quad 0 < R < q^4 - 2, \quad E \geq q^4 > R + 2, \quad k > R,$$

and, with $M_0 = D + v^2$, the inequality

$$dM_0^2 A > Q(Dv^2 + z_0^2) = e + 1.$$

These steps do not use the main root or its discarded coordinate γ. Since $M = 2M_0$, they imply

$$dM^2 A > 4(e + 1), \qquad b\sqrt{D} \leq e + 1. \tag{7}$$

The second inequality is $2vz_0\sqrt{D} \leq Dv^2 + z_0^2$. All quantities used as positive bounds are positive, with $d \geq 1$ and $D, G, A > 1$.

For proof purposes, $\mathcal{R}_{12}$ is the product over the eight choices $r = \pm\sqrt{G}$, $f = \pm\sqrt{D}$, and $\tau = \pm\sqrt{A}$ of

$$d(S - \ell r - HM\tau)^2 - H^2(e - bf). \tag{8}$$

Taking the quadratic norms first in f, then in τ, then in r gives exactly the displayed polynomial computation. Its zero supplies a real algebraic tuple satisfying

$$r^2 = G, \quad f^2 = D, \quad \tau^2 = A, \quad dy^2 = e - bf, \quad S = \ell r + H(M\tau + y). \tag{9}$$

Indeed the first three radicals are real, and the last equality defines a real y.

There is at most one such tuple for the given rational S. First, $\sqrt{A} < T_*$ and $\sqrt{D} < F_*$. Moreover

$$y^2 = \frac{Q(vf - z_0)^2 - 1}{Q - 1} < 2(v\sqrt{D} + z_0)^2, \qquad |y| < Y_*.$$

It follows that $|M\tau + y| < U_*$. Two different signs of r would therefore change ℓr by $2\ell\sqrt{G} > 2HU_*$, more than the remaining contribution can change. Thus r is unique. Next, (7) gives

$$M^2 A > \frac{e + b\sqrt{D}}{d} \geq y^2.$$

Two different signs of τ cannot give the same $M\tau + y$, because their difference $2M\sqrt{A}$ exceeds the possible difference of the y terms. Therefore τ, then y, then f are unique; the last conclusion uses $b \neq 0$.

Every algebraic embedding of the field generated by this tuple fixes S and the integer coefficients, and produces another tuple in (9). Its r, f, τ are real; the last equality forces its y to be real as well. Uniqueness fixes every generator under every embedding. Characteristic-zero separability, or the fixed field theorem in a normal closure, now implies that all four roots are rational.

The rational numbers r, f, τ have integer squares and hence are integers. The last equality in (9) makes Hy an integer. If $y = u/v_0$ is in lowest terms, then $v_0 \mid H$ and $v_0^2 \mid d$. But

$$Q = \Delta^2 H^2 i^2 c^4, \qquad \gcd(H, Q - 1) = 1.$$

Thus y is an integer. Set

$$\gamma = (r - ac - X)/H.$$

The same equality yields $t = \ell\gamma + M\tau + y$, so $\ell\gamma$ is an integer. Since $\gcd(\ell, H) = 1$, γ is an integer too. We have restored the exact equations

$$\begin{aligned}(ac + X + \gamma H)^2 - \Delta c^2 &= 1,\\ f^2 - \Delta(Hi)^2c^4 &= 1,\\ \tau^2 &= 1 + L_0(L_0 + k),\\ QV_a^2 - (Q-1)y^2 &= 1, \qquad V_a = c(Tf-1) - Rf^2.\end{aligned}$$

Here $e - bf = Q(vf - z_0)^2 - 1$ because $f^2 = D$. Also $y \neq 0$: otherwise $QV_a^2 = 1$ would contradict $Q > 1$. Taking absolute values restores positive τ, y.

The signed-core soundness argument cited above applies to the possibly signed f, γ (equivalently signed T after changing the sign of f). Its normalized rank step uses only the nonzero multiplier Hi. It recovers the main index $p = R$, the positive main-root branch, $\varepsilon = 1$, and the native masks. Thus it decodes a blank accepting K215 tableau. This use of signed-core soundness is essential: the older theorem restricted to positive γ alone would not justify this step.

5.3 Completeness with the new divisibility restriction

Take the canonical positive witnesses supplied by a blank accepting tableau. They have $R \equiv 3 \pmod 4$, $c = \psi_{a+2}(R)$ and main root $\chi_{a+2}(R)$, where $\chi_A(n) + \psi_A(n)\sqrt{A^2-1} = (A + \sqrt{A^2-1})^n$. Keep the basic coordinates, main γ, and τ. We must rechoose the strong and auxiliary witnesses; an arbitrary old witness need not have its multiplier divisible by H.

Let $N = Hc^2$. The matrix

$$\begin{pmatrix} \chi_{a+2}(R) & \Delta\psi_{a+2}(R) \\ \psi_{a+2}(R) & \chi_{a+2}(R) \end{pmatrix}$$

has determinant one. Modulo N it belongs to the finite group $\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$, whose cardinality is at most N^4. Its order divides $(N^4)!$. Consequently, for $m = R(N^4)!$, the Pell multiplication law gives $Hc^2 \mid \psi_{a+2}(m)$. Put

$$i = \frac{\psi_{a+2}(m)}{Hc^2}, \quad f = \chi_{a+2}(m), \quad \rho = \Delta\psi_{a+2}(m), \quad V_a = \frac{\chi_\rho(R)}{\rho}, \quad y = \psi_\rho(R).$$

These are positive integers; oddness of R gives integrality of V_a. The odd Chebyshev quotient identities, using $R \equiv 3 \pmod 4$, give

$$V_a \equiv -c \pmod f, \qquad V_a \equiv -R \pmod c, \qquad f^2 \equiv 1 \pmod c.$$

For clarity, their inputs are $\rho^2 \equiv -\Delta = 1 - (a+2)^2 \pmod f$ and $\rho^2 \equiv 0 \pmod c$; they do not require the old auxiliary index. Thus

$$o = (V_a + c)/f, \qquad T = (o + Rf)/c$$

are positive integers. The second divisibility follows by multiplying $of \equiv -R \pmod c$ by f and using $f^2 \equiv 1 \pmod c$. They restore exactly $V_a = c(Tf-1) - Rf^2$. Finally choose $t = \ell\gamma + M\tau + y > 0$. Then (9) holds, proving completeness.

For the single equation, a zero of (6) makes all three factors integer units. Since $\Delta \equiv 0$ or $3 \pmod 4$, we have $G \equiv 0$ or $1 \pmod 4$. Therefore $\mathcal{R}_{12} = E_0^2 - GE_1^2$ cannot be $2 \pmod 4$: the factor $1 + \mathcal{R}_{12}$ cannot be -1. It must be 1, and N_k, N_t have equal signs. The soundness proof deliberately allowed those signs, and so proves the product claim as well. Composing with the K215/refutation equivalence proves the theorem.

5.4 Degree, reproducibility, and further questions

The degrees relevant to the calculation are

quantity	c	a	Δ	L_0	A	I	D	Q	z_0	e
deg	5	6	12	11	22	17	46	58	50	158

and

$$\deg b = 114, \quad \deg M = 46, \quad \deg U_* = 57, \quad \deg \ell = 63, \quad \deg S = 74, \quad \deg W = 52.$$

For an exact certificate, replace every unknown by λ times an independent indeterminate and work with formal Laurent series at infinity. The two roots of G have expansions $r_+ = ac + 2c +$ terms of degree below 5 and $r_- = -r_+$. Thus $Z_+ = S - \ell r_+$ has degree 68, with leading form $-2\ell_{\mathrm{top}} c_{\mathrm{top}}$, and Z_- has degree 74, with leading form $2\ell_{\mathrm{top}} a_{\mathrm{top}} c_{\mathrm{top}}$. The term $W\tau$ has degree 63; the terms $H^2 e$ and $H^2 bf$ have degrees 170 and 149. Hence the leading term of each factor (8) is dZ^2, of degree 194 or 206. There are four of each. This proves

$$\deg \mathcal{R}_{12} = 4(194) + 4(206) = 1600, \qquad \deg \mathcal{P}_{12} = 1600 + 7 + 2 = 1609.$$

The leading form of $\mathcal{R}_{12}$ is the nonzero polynomial $2^{16} d_{\mathrm{top}}^8 \ell_{\mathrm{top}}^{16} a_{\mathrm{top}}^8 c_{\mathrm{top}}^{16}$, so the claim is uniform for valid fixed coefficients.

The generator emits every gate. A separate verifier compares the circuit with multiplication in the eight-dimensional algebra generated by r, f, τ on 48 signed assignments. Two modular line restrictions give degrees 1600 and 1609, agreeing with the uniform proof above. These are finite arithmetic audits; neither they nor the machine's formal endpoint constitute a proof-assistant formalization of this new reduction.

The main remaining opportunities are to lower the degree of the coprimality-enforcing multiplier restriction, to replace the three norms by a less expensive elimination, and to combine the twelve-variable construction with the sharper signed-domain analysis. Formalizing the branch separation and denominator argument would isolate a small reusable lemma from the longer inherited native compiler proof.

6 A direct polynomial in twenty-seven unrestricted integers

The preceding nine-variable construction minimizes the number of nonnegative unknowns among the constructions supplied here, but it has a very large degree. We now give a direct signed-coordinate alternative. Combining the weighted three-root elimination of package 09 with the signed-core argument of package 14 reduces that direct family's unrestricted-integer count from 29 to 27. Two versions have exact degrees 1004 and 1116, with 137 and 136 arithmetic operations, respectively. The reduction below is an ordinary mathematical proof; the new specialized circuits have not been formalized in Lean or Rocq.

Throughout this section, (β, K, M_C, M_F) is the complete four-port T4 coefficient tuple defined earlier, with $\beta = B - 1$ and with M_F the *shifted* mask. The proof uses the established native compiler interface and its canonical padding by independent powers of five. The complete older package-06 tuple is an alternative binding with the same properties; individual ports from different tuples must not be combined.

6.1 The polynomial and its domain

Temporarily retain seven positive coordinates

$$J, F, \alpha, z, s, \eta, \zeta > 0$$

and six unrestricted integer coordinates

$$h, r, T, u, \gamma, t \in \mathbb{Z}.$$

Set $i = 2r + 1$ and $w = u^2$, and use the following polynomial abbreviations:

$$\begin{aligned} &q = \beta J + 1, && X = wq, && Y = sq^3, && E = XY, && k = \eta + \zeta, \\ &c = kY + \eta, && a = E + Y, && H = 4a + 3, && \Delta = a^2 + H. \end{aligned}$$

The main root, packed index, and three outer factors are

$$\begin{aligned} D_{\text{main}} &= X + ac + \gamma H, \\ R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \\ N_{\text{main}} &= D_{\text{main}}^2 - \Delta c^2, \\ N_k &= k - hE - R, \\ N_t &= (K + w)(q - F - \alpha) + q - F - z(q-1). \end{aligned}$$

For the eliminated roots define

$$\begin{aligned} &L_0 = EkY, && A = 1 + L_0(L_0 + k), && I = ic^2, \\ &D = 1 + \Delta I^2, && Q = \Delta(D-1), && d = Q - 1, \\ &v = cT, && z_0 = c + RD, && W_0 = Dv^2 + z_0^2, \\ &e = QW_0 - 1, && B_0 = 2Qvz_0. \end{aligned}$$

The two admissible weights are

$$M_{\text{full}} = D + v^2, \qquad M_{\text{low}} = kI^2 + T^2.$$

For either choice M put

$$\begin{aligned} C &= d(t^2 + M^2 A) - e, \\ U &= C^2 + DB_0^2 - 4d^2t^2M^2A, \\ \mathcal{R} &= U^2 - D(2CB_0)^2, \qquad (10) \\ P_{27} &= N_{\text{main}} N_k N_t (1 + \mathcal{R}) - 1. \qquad (11) \end{aligned}$$

All displayed symbols are substituted polynomials, not further quantified unknowns. The alternative four-equation system is

$$N_{\text{main}} = N_k = N_t = 1, \qquad \mathcal{R} = 0. \qquad (12)$$

To obtain literally 27 unrestricted unknowns, replace each of the seven positive coordinates x by an independent block

$$x = \text{Pos}(a_x, b_x, c_x) = a_x^2 + a_x + b_x^2 + c_x^2 + 1. \qquad (13)$$

This is positive for every integer triple and represents every positive integer. Indeed, $4x - 3$ is positive and congruent to 1 modulo 4, so Legendre's three-square theorem represents it as a sum of three squares. Exactly one of those squares is odd and two are even; writing the three roots as $2a_x + 1, 2b_x, 2c_x$ gives (13). Thus the total is $7 \cdot 3 + 6 = 27$, with no extra variable for a sign or a denominator.

Claim. For either weight, (11) after (13), and likewise (12), has an integer solution if and only if K215 halts on the blank tape. Consequently the absence of integer solutions is equivalent in ZFC to Con(ZFC).

6.2 Unit extraction and bounds before restoring roots

At a zero of P_{27}, all four factors are integer units. Since $\Delta \equiv 0$ or $3 \pmod 4$, the main norm cannot be -1. Moreover $\mathcal{R} \equiv U^2 \pmod 4$, so $1+\mathcal{R}$ cannot be -1. Therefore

$$N_{\text{main}} = 1, \quad \mathcal{R} = 0, \quad N_k = \epsilon, \quad N_t = \nu, \quad \epsilon, \nu \in \{-1, 1\}, \quad \epsilon\nu = 1. \tag{14}$$

The system is the special case $\epsilon = \nu = 1$.

Initially $X \geq 0$, while Y, k, c, a, H, Δ are positive. If $u = 0$, then $X = E = 0$, $a = Y$, and

$$N_{\text{main}} = H(2ac\gamma + \gamma^2 H - c^2).$$

This cannot equal 1 because $H = 4Y + 3 > 1$. Thus $u \neq 0$, $w \geq 1$, and $X \geq q \geq 16$. The inherited weak-transport argument and exact mask bounds give

$$q - F - \alpha \geq 0, \qquad q(q^2-1) < R < q^4 - 2, \qquad E \geq q^4 > R + 2, \qquad a > R + 2. \tag{15}$$

These bounds use only the transport/index units and the coefficient interface, not any of the deleted roots. A negative h would give $k = R + \epsilon + hE < 0$, so $h \geq 0$. Also $Y \geq q^3 \geq 4096$ and $c < k(Y+1)$.

6.3 The main norm already forces a positive index quotient

This additional observation is needed to use the lower weight without assuming the deleted first norm. Let $A_0 = a + 2$, so $\Delta = A_0^2 - 1$. Write the standard Pell sequences as

$$\chi_{A_0}(n) + \psi_{A_0}(n)\sqrt{A_0^2 - 1} = (A_0 + \sqrt{A_0^2 - 1})^n.$$

The main norm gives $c = \psi_{A_0}(p)$ and $|D_{\text{main}}| = \chi_{A_0}(p)$ for $p \geq 1$. Since $c > 1$, $p \neq 1$.

If $p = 2$ and $D_{\text{main}} > 0$, then $D_{\text{main}} - ac = H + 4$. Its defining congruence would give $X \equiv 4 \pmod H$, impossible because $16 \leq X < H$. If $p = 2$ and $D_{\text{main}} < 0$, direct expansion gives $4(D_{\text{main}} - ac) \equiv -1 \pmod H$. This would give $4X \equiv -1 \pmod H$, impossible because $0 < 4X + 1 < H$. Thus $p \geq 3$ and

$$c \geq \psi_{A_0}(3) = 4A_0^2 - 1.$$

If $k \leq E$, then

$$c < k(Y+1) \leq E(Y+1) \leq E(E+1) < 4(E+2)^2 - 1 < 4A_0^2 - 1,$$

a contradiction. Hence $k > E$. In view of (15), $h = 0$ would imply $k = R + \epsilon < E$. We have therefore proved

$$h \geq 1, \qquad k > E, \qquad k > R \tag{16}$$

before restoring the first Pell root.

6.4 Strict domination for both weights

We establish

$$AM^2 > 4W_0, \qquad dM^2A > 2QW_0. \tag{17}$$

The second follows from the first because $Q \geq 2$ and $d = Q - 1 \geq Q/2$.

For the full weight, (16) implies

$$c^2 + R^2 < k^2((Y+1)^2 + 1) \leq 5k^2Y^2, \qquad A > L_0^2 = k^2X^2Y^4 > 16(c^2 + R^2).$$

Here $X \geq 16$ and $Y \geq 1$ give the last strict inequality. Since $M^2 \geq 4Dv^2$, $M^2 \geq D^2$ and $z_0^2 \leq 2D^2(c^2 + R^2)$,

$$4W_0 \leq M^2(1 + 8(c^2 + R^2)) < AM^2.$$

For the lower weight, $A_0 < 2E$, $c < 2kY$ and $L_0 = kEY$ give

$$A > \frac{A_0^2 c^2}{16}, \qquad A > \frac{A_0^4 Y^2}{16} > 32A_0^4. \tag{18}$$

The second uses $k > E$, already proved, and $Y \geq 4096$. Furthermore $I^2 = i^2c^4 > c$, so

$$D < A_0^2 I^2, \qquad z_0 = c + RD < 2kA_0^2 I^2.$$

The inequalities $M \geq kI^2$ and $M^2 \geq 4kI^2T^2$, together with $k > R > 3q \geq 48$ and the first estimate in (18), imply $AM^2 > 8Dv^2$; when $T = 0$ this is immediate. The second estimate gives $AM^2 > 8z_0^2$. Adding and dividing by two proves (17). In particular, for every integer t,

$$C = dt^2 + dM^2A - (QW_0 - 1) > dt^2 + QW_0 + 1 > 0. \tag{19}$$

6.5 Excluding the two new exceptional branches

The original positive-variable elimination assumed $t, T > 0$. Here they are unrestricted, so their possible zeros require proof.

The branch $t = 0$. At $t = 0$, the resultant is

$$\mathcal{R} = (C^2 - DB_0^2)^2.$$

But $DB_0^2 = 4Q^2Dv^2z_0^2 \leq Q^2W_0^2$, whereas (19) gives $C > QW_0 + 1$. Thus $\mathcal{R} > 0$, contradicting (14). Hence $t \neq 0$.

The branch $T = 0$. First observe that

$$2 < z_0 < Q - 1. \tag{20}$$

The lower bound is immediate, and, because $\Delta > R$,

$$\begin{aligned} Q - z_0 &= (\Delta - R)D - \Delta - c \\ &\geq D - \Delta - c = 1 + \Delta(i^2c^4 - 1) - c > 1. \end{aligned}$$

If $T = 0$, then $v = B_0 = 0$, and $\mathcal{R} = 0$ gives $C^2 = 4d^2t^2M^2A$. Since d, t, M are nonzero, $\tau = C/(2dtM)$ is rational with square $A \in \mathbb{Z}$ and hence is an integer. Put $y = t - M\tau$. The definition of C gives $dy^2 = e = Qz_0^2 - 1$, so

$$(Q - 1)(y^2 - z_0^2) = z_0^2 - 1 > 0.$$

Integrality implies $|y| \geq z_0 + 1$. Using (20) then yields the contradiction

$$z_0^2 - 1 \geq (Q - 1)(2z_0 + 1) > z_0(2z_0 + 1) > z_0^2 - 1.$$

Therefore $T \neq 0$, and hence $vB_0 \neq 0$. Every denominator in the following reconstruction has now been proved nonzero.

6.6 Restoring the signed source equations

Define the proof-only rational numbers

$$f = -\frac{U}{2CB_0}, \qquad \tau = \frac{C + B_0 f}{2dtM}, \qquad y = t - M\tau.$$

Equation $\mathcal{R} = 0$ gives $f^2 = D$. The definition of U then gives $(C + B_0 f)^2 = 4d^2t^2M^2A$, hence $\tau^2 = A$. A rational number whose square is an integer is integral, as follows by reducing its fraction to lowest terms. Thus f, τ, y are integers. Expansion gives

$$dy^2 = e - B_0 f = Q(vf - z_0)^2 - 1.$$

Since $f^2 = D$, the expression $vf - z_0$ is exactly

$$V = c(Tf - 1) - Rf^2.$$

We have restored all three normalized equations

$$\tau^2 - L_0(L_0 + k) = 1, \qquad f^2 - \Delta i^2 c^4 = 1, \qquad QV^2 - (Q-1)y^2 = 1. \tag{21}$$

Together with (14), these are exactly the normalized source product equations. The recovered f, τ, y may have signs.

We now reuse package 14's *signed-core soundness argument*: its seven positive ports are retained, the multiplier $i = 2r + 1$ is nonzero, the square scale $w = u^2$ has already been proved positive, and all the source equations have been restored. That argument classifies absolute Pell roots, uses the two auxiliary congruences to recover the exact main index $p = R$, excludes the negative main-root branch by its short residue interval, and recovers the native masks and the blank tableau. It allows f, h, T, τ, y, γ to be signed; it does not infer soundness by evaluating the positive-domain theorem at a signed tuple. In particular, it forces the two remaining unit signs to be positive. The result is a blank halting run of K215. This imported argument is invoked only after (21), never to establish the reconstruction denominators.

6.7 Completeness, including the T4 coefficient binding

The canonical compiler witness has $R \equiv 3 \pmod 4$ and $c = \psi_{a+2}(R)$ odd. As in package 14, choose the strong index $m = cR$ and put

$$f = \chi_{a+2}(m), \qquad i = \frac{\psi_{a+2}(m)}{c^2}.$$

The expansion of $(\chi_{a+2}(R) + c\sqrt{\Delta})^c$ shows $c^2 \mid \psi_{a+2}(m)$. Since m is odd, both the numerator and c^2 are odd, so i is a positive odd integer. The standard odd quotient identities at $R \equiv 3 \pmod 4$ give the required positive auxiliary witnesses and the exact polynomial V.

The canonical spatial and temporal dimensions may independently be sufficiently large powers of five. Their product N is odd. For T4, $B = 2^{d_0}$ with $d_0 = 5^{18}$ odd, so

$$q = 2^{d_0 N}, \qquad w = 2^{R - d_0 N}$$

is a positive square: $R - d_0 N$ is positive and even. This is the precise reason that package 14's square-scale completeness survives the T4 replacement. The whole tuple has the same compiler interface; no port is changed in isolation. The older package-06 value $d_0 = 5^{67}$ has the same parity property.

Choose integers r, u with $i = 2r + 1$ and $w = u^2$. For either weight set $t = M\tau + y$. The forward identities give $\mathcal{R} = 0$, while the three outer factors equal 1. Finally use the surjectivity of (13) to choose the seven independent integer triples. This proves completeness for the literal 27-integer polynomial and system. The stated equivalence with consistency now follows from the earlier K215 calibration; it is not a proof of consistency itself.

6.8 Exact degrees, circuit lengths, and evidence

Weight and form	Variables	Equations	Exact degree	Operations
Full product	27 in $\mathbb{Z}$	1	1116	136
Lower product	27 in $\mathbb{Z}$	1	1004	137
Full system	27 in $\mathbb{Z}$	4	1064 maximum	130
Lower system	27 in $\mathbb{Z}$	4	952 maximum	131

The source products have 84 and 85 operations. Each of seven positivity substitutions costs seven, the odd multiplier costs two, and the square scale costs one; the increment is therefore 52. Every binary addition, subtraction and multiplication is counted, including a product by a fixed numeral. The lower product has 73 multiplications and 64 additions or subtractions. Removing the six finalizer gates gives the corresponding systems.

For an exact degree proof, assign weight two to the seven positive source ports and w, and weight one to i, h, T, γ, t. The relevant degrees are

expression	q	X	Y	E	k	c	a	Δ	R
degree	2	4	8	12	2	10	12	24	8

and

expression	A	I	D	Q	v	z_0	e
degree	44	21	66	90	11	74	238.

The outer factor degrees are $35, 13, 4$. For the lower weight, M has degree 44, and the unique leading term of C is $-e_{\mathrm{top}}$. The unique leaders of U and $\mathcal{R}$ are then C_{top}^2 and U_{top}^2, giving degrees $238, 476, 952$; the competing last norm term has degree 892. Hence the product degree is $952 + 35 + 13 + 4 = 1004$. For the full weight, M has degree 66, and dM^2A dominates C. The successive degrees are $266, 532, 1064$, giving the product degree 1116.

There is no hidden cancellation of the outer leaders. They are respectively

$$8\gamma a_{\mathrm{top}}^2 c_{\mathrm{top}}, \qquad -hE_{\mathrm{top}}, \qquad w(q_{\mathrm{top}} - F_{\mathrm{top}} - \alpha_{\mathrm{top}}) - z_{\mathrm{top}} q_{\mathrm{top}}.$$

Each is nonzero, and their product with the resultant leader is nonzero. The top parts of (13), $i = 2r+1$, and $w = u^2$ preserve nonvanishing: set the second and third variable of every positivity block to zero, leaving an injective substitution into distinct square variables and the nonzero scalar substitution $i \mapsto 2r$. These degrees are exact for every valid fixed $\beta > 0$.

The supplied generator emits all four literal circuits, checks operation counts and complete liveness, and compares both weights with an independent formula on 96 signed assignments each. It also checks 144 isolated signed root reconstructions, 6786 finite instances of the gap inequality, and two modular degree diagnostics per circuit. Both ordinary and optimized Python runs passed. These are finite algebraic checks; the universal proof is the argument above, and the inherited compiler and K215 calibration remain explicit mathematical dependencies.

7 The fixed T4 coefficient recipe

This section freezes one complete instance of the compiler. It reproduces the bounded definition proved by package 11 [5], with separate notation for the native and shifted masks. The full ordered tile list, literal K215 transitions, predicate and exact coefficient metadata are included in the archive. Neither the tile enumeration nor the coefficient definition asks whether K215 halts.

7.1 The finite local language

There are 439 cell symbols. Headless tape bits have codes $0, 1$; a bit $b \in \{0, 1\}$ with head state $q \in \{0, \ldots, 215\}$ has code $2(q + 1) + b$. The special cells are

$$L = 434, \quad Q = 435, \quad R = 436, \quad V = 437, \quad H = 438.$$

Here L, Q, R are initialized phases with payloads blank/headless, blank/state-0, blank/headless. V is a vertical boundary and H a horizontal boundary. They carry no tape payload. Only the completed halt transition stays in place: $(215, b) \mapsto (b, 0, 215)$. The actual first K215 action is $(0, 0) \mapsto (1, +1, 93)$; the transition entering halt retains its actual write and movement.

A tile is a quadruple (l, c, r, n) of left, center, right and next-time center cells. The exact predicate is the supplied `upstream/t4/bridge/check_t4.py`. In mathematical terms:

1. At most one of l, c, r has a head; the initialized phase Q counts as a head.
2. c is V exactly when n is V. Adjacent c, r cannot both be V, and c, n cannot both be H.
3. If neither c nor r is V, they agree on whether they are H. For a center V, there is no further update requirement.
4. A center H has next-time center in $\{L, Q, R\}$. For an initialized interior center, the left boundary requires L, the right boundary requires R, and away from a right boundary the phase edge is one of LL, LQ, QR, RR.
5. An interior head may not move into an adjacent V. If the next center is H, the current center is headless or has halt state 215.
6. Otherwise the next center is ordinary, with no initialized phase, and is the exact three-neighbor deterministic successor. A neighbor V is blank/headless; a neighbor H cannot supply this update. The new bit is the center's written or stable bit, and incoming heads are given by the completed transition table. Multiple arrivals are forbidden.

Place $S = (434, 435, 436, 1)$ first and $E = (438, 0, 0, 438)$ second, then list the other passing tuples lexicographically. Both distinguished tiles pass the local predicate. The second cannot occur in a valid plane: its left H would have a non-boundary, non-H right neighbor, contradicting horizontal boundary propagation. Forbidding this tile therefore excludes no genuine computation.

There are exactly 37 075 tiles. The independent count by center class is

Center class	Tiles	Center class	Tiles
Headless ordinary 0	9,148	Headless ordinary 1	9,148
Working-head symbols	8,600	Halt-head symbols	110
L	888	Q	4
R	1,750	V	4,793
H	2,634	Total	37,075

The ordered list has SHA-256

`9c2e473b5dbc5b3dc1468a31aefdbec6a901d6ba1e6b1ec9212698d41e1d84b1.`

For a cyclic sequence of tiles w_i with stride h, the three overlaps are $w_i(r) = w_{i-1}(c)$, $w_i(c) = w_{i-1}(l)$ and $w_i(n) = w_{i-h}(c)$. With $a_i = w_i(c)$ this gives $w_i = (a_{i+1}, a_i, a_{i-1}, a_{i-h})$, hence a valid pullback plane a_{-x-ht}. This is why residue collisions and arbitrary strides do not invalidate soundness. The marked initialized row lies between vertical boundaries and the following horizontal boundary requires a halted last interior head. For completeness, sufficiently wide blank margins make the added one-head filter harmless, including helical seams. Independent width and height powers of five remain available.

7.2 Literal coefficient formulas

Put

$$A = 2^{16}, \quad \rho = 2^{390625}, \quad B = \rho^{9765625} = 2^{5^{18}}, \qquad \mu = A - 2 + \sum_{j=1}^{18743} A^j.$$

For $0 \le u \le 438$, use copy positions

$$e_l(u) = 40587 + u, \quad e_c(u) = 40148 + u, \quad e_r(u) = 39709 + u, \quad e_n(u) = 38831 + u.$$

Let $\mathcal{E}$ contain selector positions $0, \dots, 37074$, these 1756 copy positions, the dummy 42343, and the four anchors $43661, 130983, 392949, 1178847$. Its size is 38 836. For the ordered tile $w_i = (l_i, c_i, r_i, n_i)$ define

$$v_i = 1 + A^{1+l_i} + A^{440+c_i} + A^{879+r_i} + A^{1318+n_i} + \sum_{t=1}^{4} A^{1756+t},$$

$$v_{e_l(u)} = A^{1+u}, \qquad v_{e_c(u)} = A^{440+u},$$
$$v_{e_r(u)} = A^{879+u}, \qquad v_{e_n(u)} = A^{1318+u},$$
$$v_{3^{t-1}43661} = A^{1756+t} \quad (1 \le t \le 4), \qquad v_{42343} = 0.$$

Then

$$\begin{aligned}
D_C &= \rho^{1317} + \rho^{2228468} + \rho^{349288} + \rho^{1047864} \\
&\quad + \sum_{e \in \mathcal{E}} v_e (\rho^{4586163-e} + \rho^{6943861-e}), \\
D_R &= \rho^{2228029}, \\
M_F^0 &= \mu(\rho^{4586163} + \rho^{6943861}) + \sum_{u=0}^{438} \rho^{40148+u} \\
&\quad + \sum_{u=0}^{438} (\rho^{2228029+40587+u} + \rho^{2228029+40148+u}) + \rho^{392949} + \rho^{1178847}, \\
M_C &= B - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \ne 1}} \rho^e - 2\rho^{42343}, \\
M_F^{\text{native}} &= M_F^0 + 4.
\end{aligned}$$

The fixed ports used in every final equation are

$$\boxed{\beta = B - 1, \quad K_0 = D_C + BD_R, \quad M_C, \quad M_F = M_F^0 + 4 + B - 1.} \tag{22}$$

These are positive integers satisfying the compiler's mask and size conditions. Package 11 proves the support geometry, carry bounds, population census and five-adic index adjustment. Our replayed finite checks verify their finite data; the arbitrary-length coding proof remains the imported mathematical theorem.

8 Verification, reproducibility and scope

8.1 Executed checks

The supplied Python programs use the standard library. The principal checks performed for this report are:

- Exact sparse construction of the 23 shifted residuals and their sum of squares, including degree four, the unit normalization residual, 346 coefficient records and the twenty-term index bound. A second independent implementation agrees on all coefficient and bound records.
- Direct T4 enumeration and independent class-factorized counting, agreeing on all 37 075 ordered tiles, plus the support, carry and population checks. The direct enumeration examines 20 624 220 quadruples after the horizontal head filter.
- Exact polynomial identities for the four-factor nine-variable norm and its unconditional majorant, small complete-homogeneous coding identities, the factorial-weighted coding identity, and 432 constructed relation-combiner witnesses.
- Sixty-four modular comparisons of generated nine-variable circuits with separate direct formulas on smaller coding dimensions, followed by 24 comparisons using the actual fixed T4 and index coefficients. The latter also compare 840 named intermediate values. Geometric sums are evaluated without division, including at noninvertible radices.
- A symbolic leading-coefficient pass through both nine-variable circuits, with all 348 named fixed leaves treated as independent symbols. The final leading coefficient is a nonzero scalar, so the degree claim survives the authentic coefficient specialization.
- Forty-eight comparisons of the twelve-variable circuit with an independent eight-dimensional radical algebra, structural gate checks, and two modular line-degree diagnostics. Its exact degree additionally has the uniform Laurent-series proof in the mathematical section.
- For each signed weight, 96 direct-formula comparisons, together with 144 isolated inverse-component checks, 6786 finite gap-obstruction checks and circuit/degree audits. The signed proof establishes the inequalities universally; these finite checks are supporting evidence.

For independent arithmetic fingerprints, the actual coefficient residues and index residues are

Modulus	β	K_0	M_C	M_F	$z_\star$	$y_\star$
101	9	12	28	32	49	12
1009	1	1000	520	356	409	696
1000003	160640	704240	343080	140964	259269	170996

For these prime moduli, $s_\star$ is first evaluated modulo $p-1$ using the actual ports modulo $p-1$, and Fermat's theorem then gives $z_\star \bmod p$. This is valid because p is odd and 2 is a unit. No Euler/Fermat exponent reduction is silently used for a nonunit. The bounded digit-polynomial evaluation then gives $y_\star \bmod p$. These are checks of the literal constant definition, not a search for a consistency witness or evidence of nonsolvability.

8.2 Archive organization and replay

The root `README.md` gives the command sequence and metric conventions. `report.tex`, the three included section files and `report.pdf` form the complete paper. The directories have these roles:

Directory	Contents
`base/`	Exact source quartic; independently generated normalized coefficients and index bound.
`coefficients/`	Ordered T4 tiles and the frozen compiler coefficient recipe.
`code/`	Independent base constructor, actual fixed-port evaluator and replay support.
`nine/`	Two literal arithmetic DAGs, constructor, identity/degree checks and actual-instance modular verifier.
`direct12/`	Twelve-variable circuit, generator, independent checks and full proof.
`integer27/`	Both signed weights, product/system circuits, exact ledgers and full proof.
`proofs/`	Source-interface audit, independent review and comparison notes.
`upstream/`	Selected pinned mathematical dependencies, T4 checkers, machine data, notices and provenance.
`verification/`	Current-session receipts and the integrated replay record.

From the extracted package, run `python3 reproduce.py` for the arithmetic and binding checks; `python3 reproduce.py -full-t4` also re-enumerates the local language. Run `make pdf` for two pdfLaTeX passes. No network access, giant coefficient expansion or proof-assistant installation is needed for these finite checks. The SHA-256 manifest records checksums for the delivered files; it is not a mathematical proof certificate. Selected upstream source references retain their historical commit metadata, while the acquisition and comparison pin of this package is the full hash on the title page.

8.3 Logical limits and next proof work

No equation is asserted to have or lack a solution independently of the requested equivalence. Under consistency, Gödel's second incompleteness theorem prevents ZFC from proving the no-solution claim. An assertion of two-sided independence would require additional care: mere consistency does not generally rule out a theory proving its own inconsistency. No stronger independence conclusion is used here.

The most useful next formalization tasks are concrete: bind the 32-variable sparse polynomial and its coefficient bound to `Jones1982.Nine.wset_iff_nine`; formalize the unconditional majorant and two-radical finalizer; encode the twelve-variable uniqueness and integrality lemmas; and formalize the signed denominator exclusions. The finite coefficient adapter also needs a proof-assistant interface before the entire new Diophantine endpoint can be called kernel checked.

For further simplification, the degree/circuit cost of the two separating weights in the twelve-variable proof is a plausible target. Another target is a smaller normalized quartic, because the Jones coding exponent grows as $5^{\nu+1}$. Neither observation is a claimed result. The supplied equations, coefficient definitions and proofs are the concrete deliverable.

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